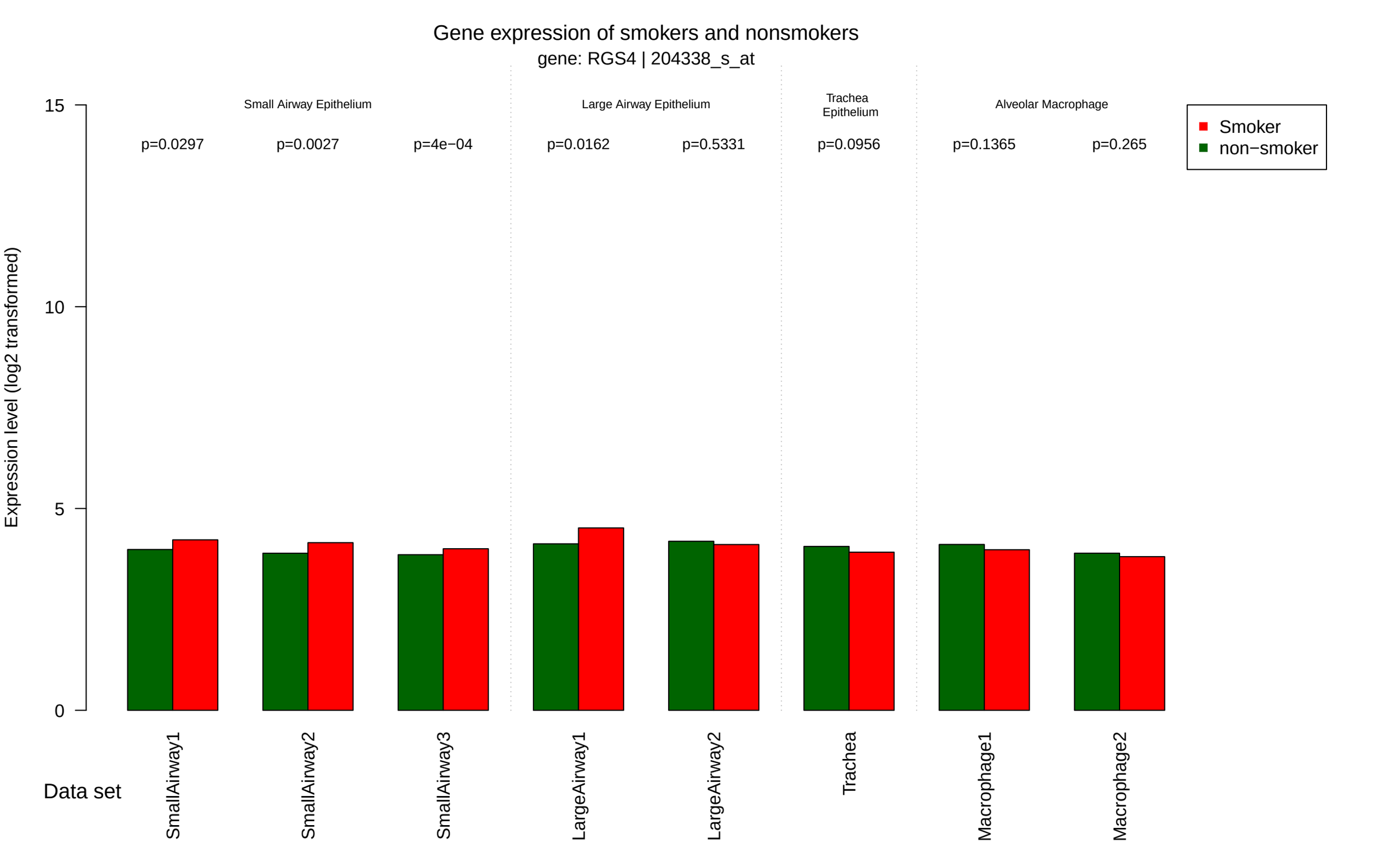
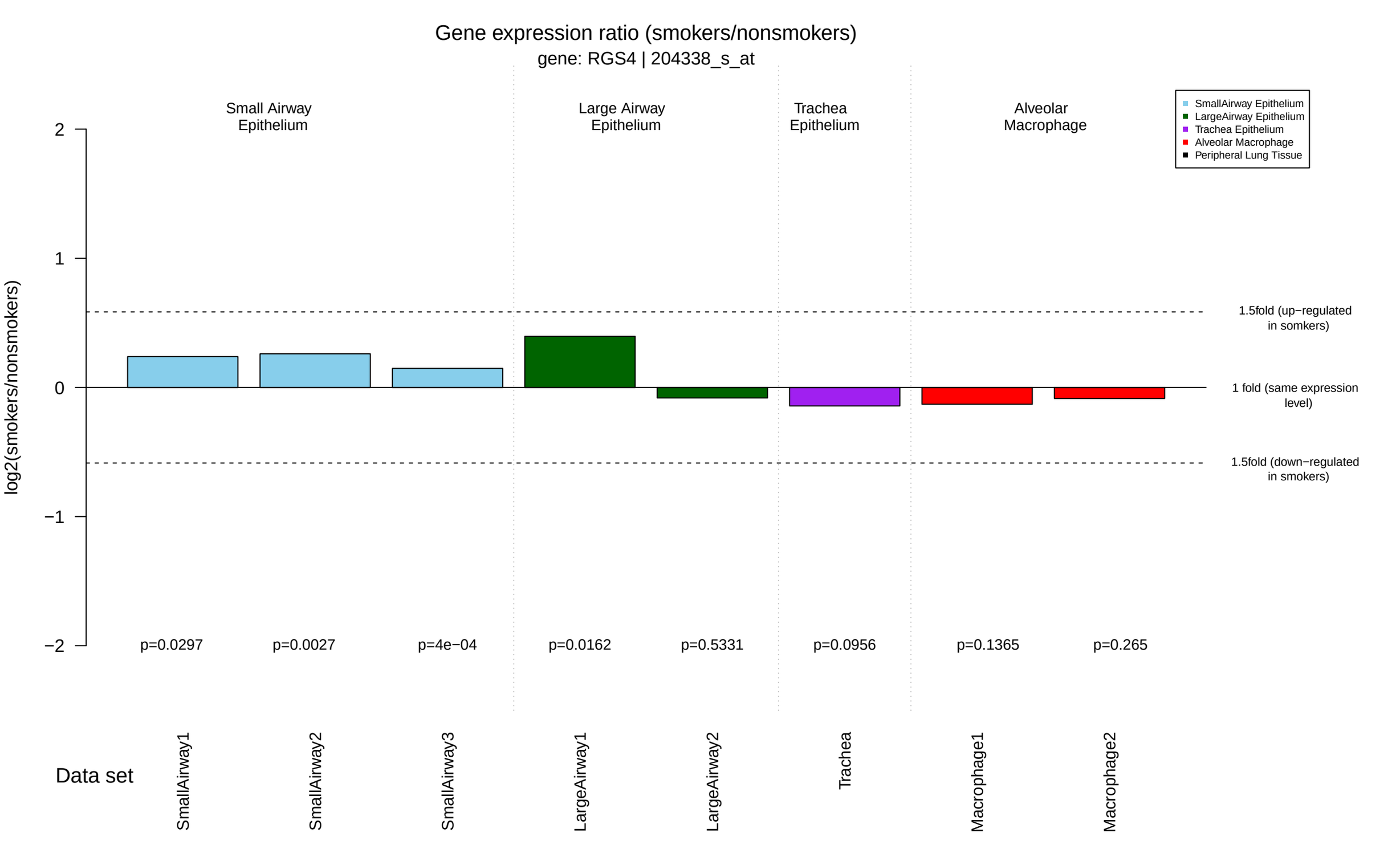
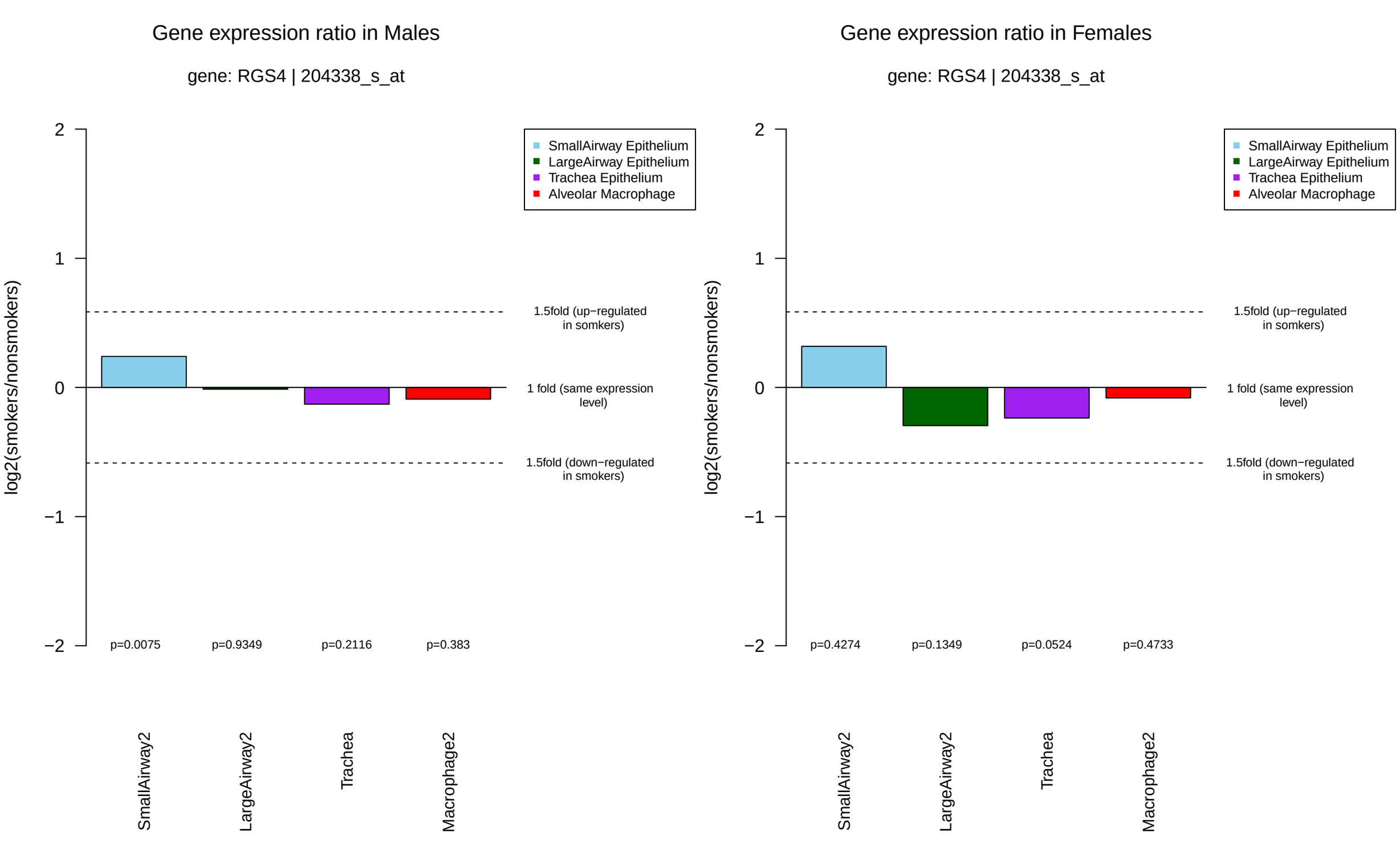
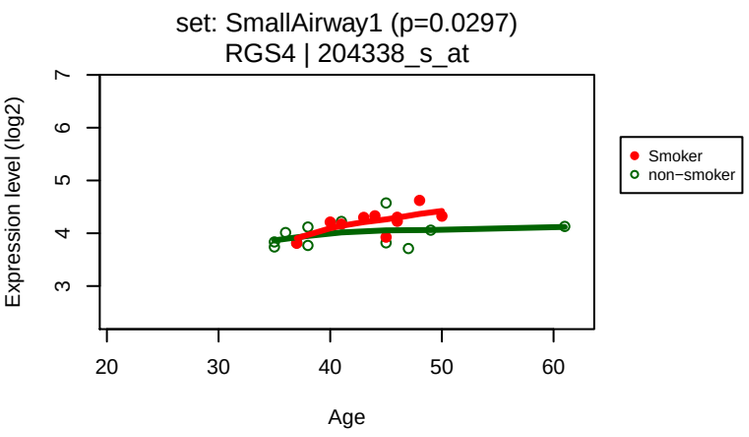
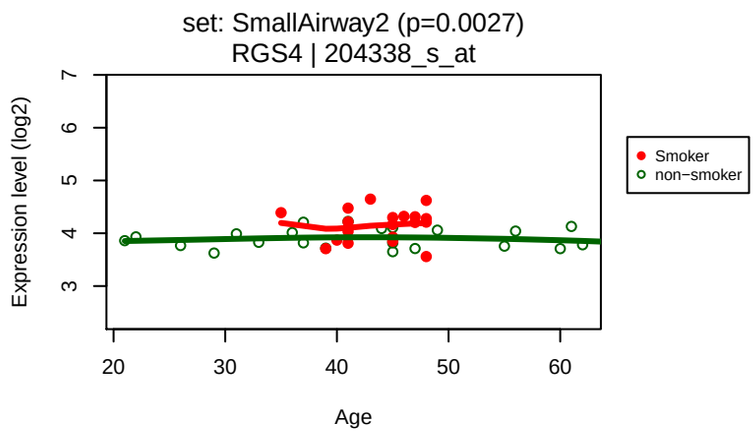
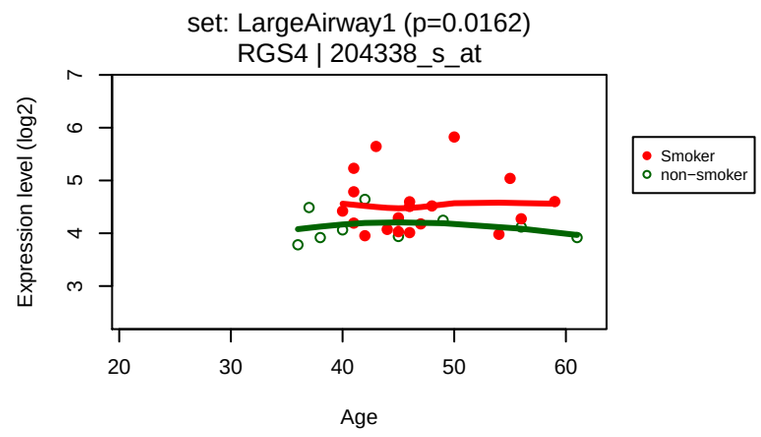

Gene expression of smokers and nonsmokers
gene: RGS4 | 204338_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0297
p=0.0027
p=4e−04
p=0.0162
p=0.5331
p=0.0956
p=0.1365
p=0.265
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

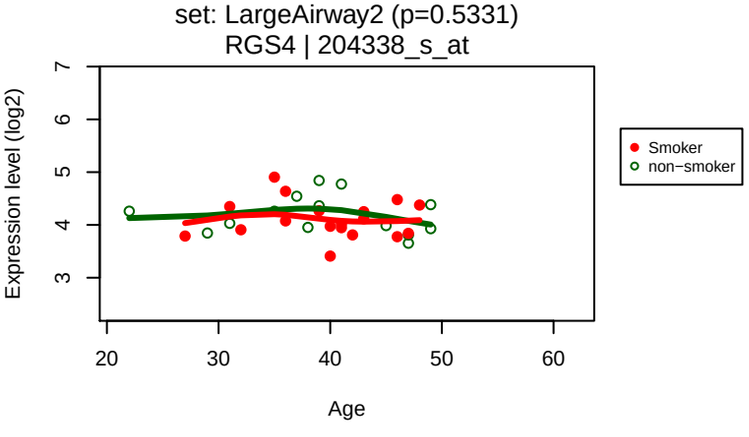

Gene expression ratio (smokers/nonsmokers)
gene: RGS4 | 204338_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0297
p=0.0027
p=4e−04
p=0.0162
p=0.5331
p=0.0956
p=0.1365
p=0.265
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

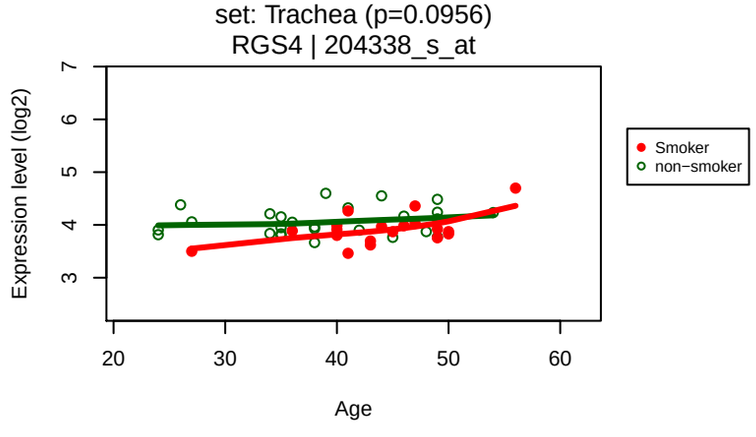

Gene expression ratio in Males
gene: RGS4 | 204338_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0075
p=0.9349
p=0.2116
p=0.383
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS4 | 204338_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4274
p=0.1349
p=0.0524
p=0.4733
SmallAirway2
LargeAirway2
Trachea
Macrophage2

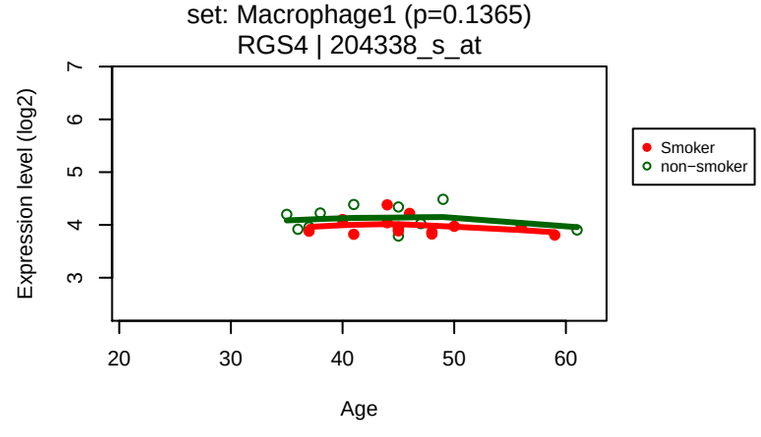

set: SmallAirway1 (p=0.0297)
RGS4 | 204338_s_at
Expression level (log2)
Age
Smoker
non−smoker

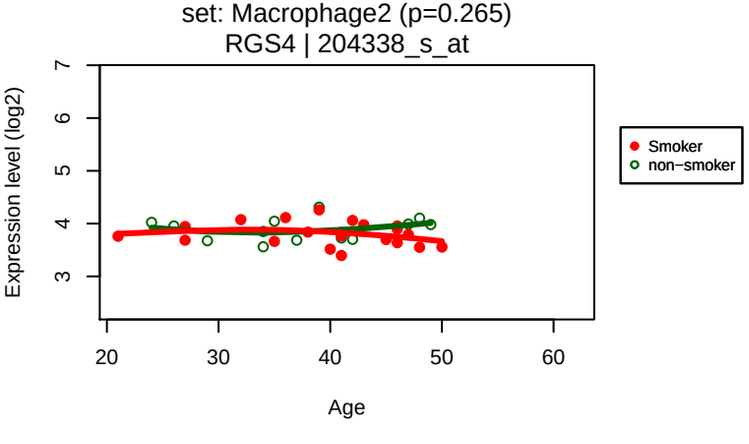

set: SmallAirway2 (p=0.0027)
RGS4 | 204338_s_at
Expression level (log2)
Age
Smoker
non−smoker

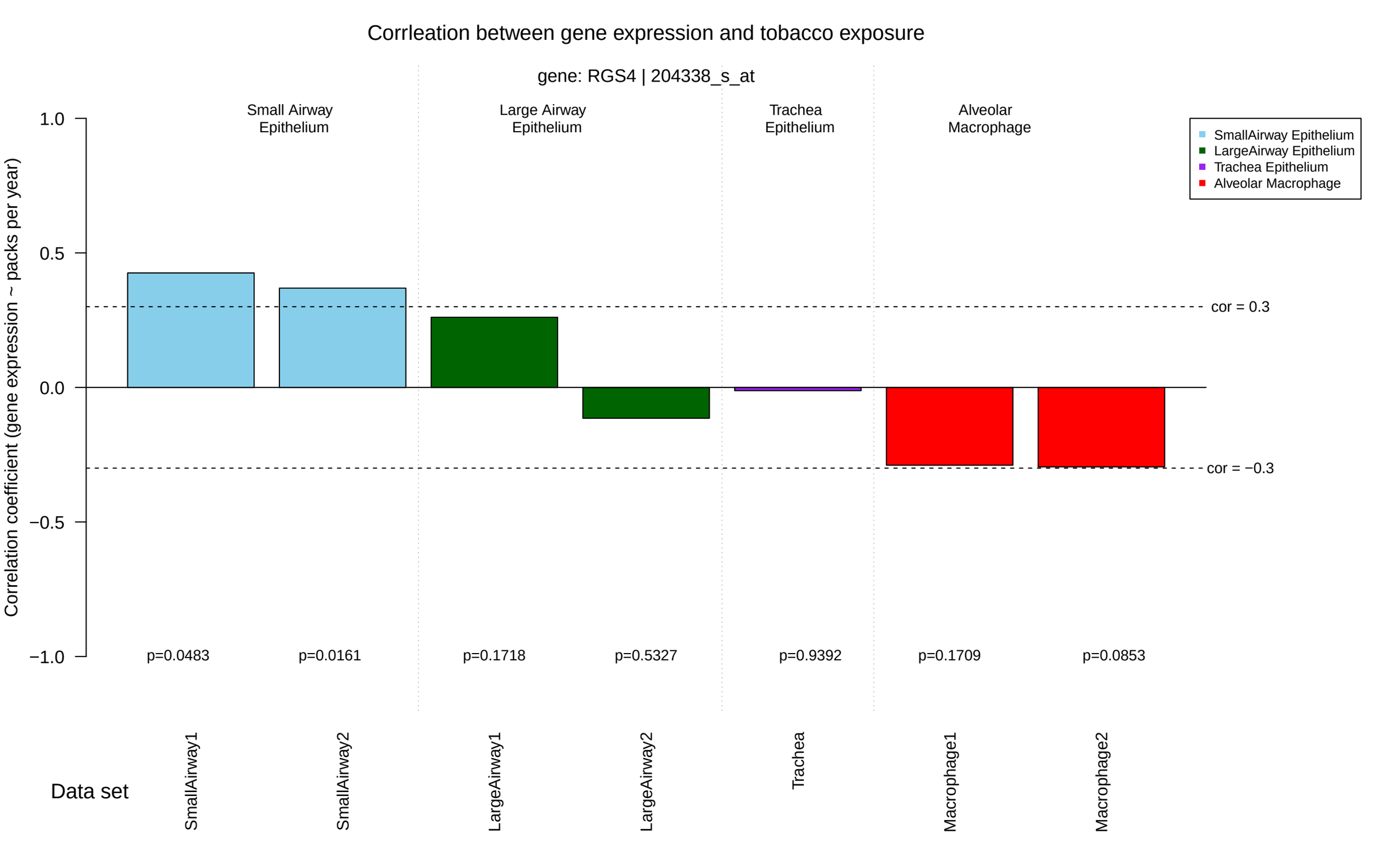

set: LargeAirway1 (p=0.0162)
RGS4 | 204338_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5331)
RGS4 | 204338_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0956)
RGS4 | 204338_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1365)
RGS4 | 204338_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.265)
RGS4 | 204338_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS4 | 204338_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0483
p=0.0161
p=0.1718
p=0.5327
p=0.9392
p=0.1709
p=0.0853
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2